BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 125 · Debugging Syntax and Logic Errors

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. For valid ages 13–18, identify the two valid boundary values.

Q2. What is an IDE's editor used for?
A. Writing source code
B. Fixing physical CPU faults
C. Managing user access rights
D. Defragmenting a hard drive

Q3. “The first index in every OCR ERL array is 1.” Correct this. Explain why.

Q4. A ticket program checks ages. “A few successful tests prove it has no faults.” Correct this. How should it be tested?

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I Do · Read new knowledge

Key vocabulary

Syntax error — Code breaks language grammar and may fail to run.

Logic error — Code runs but behaves incorrectly.

Trace and isolate — Follow values and conditions to locate the first incorrect change.

Repair and retest — Correct the cause, then check both the original and adjacent boundary cases.

New learning

Syntax error: Code breaks language grammar and may fail to run.

Logic error: Code runs but behaves incorrectly.

Trace and isolate: Follow values and conditions to locate the first incorrect change.

Repair and retest: Correct the cause, then check both the original and adjacent boundary cases.

Worked example

A loop uses total = total - value when calculating a sum. Diagnose and correct it.

1. The program runs, so it may be a logic error.

2. Trace a small input to observe subtraction.

3. Replace subtraction with total = total + value.

4. Retest several inputs and compare totals with expected sums.

We Do · Practise with guidance

Guided 1

A range check uses OR instead of AND for score >=0 and score <=100. Fix it.

Worked method: Retest -1, 0, 100 and 101.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the Boolean operator that makes the range test accept every number.

```
octet = 300
if octet >= 0 OR octet <= 255 then
    print("Valid")
else
    print("Invalid")
endif
```

iterate

Question 1

Differentiate a syntax error from a logic error. [2 marks]

Question 2

Identify the fault in if age >= 13 OR age <= 18 for acceptance age 13–18. [2 marks]

Question 3

Explain why testing after debugging is necessary. [2 marks]

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A logic error can occur in a program that runs without a syntax error.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 13 and 18 lie on the edges of the valid range.

Q2. A — Writing source code

Q3. A fixed-size indexed collection of related elements; OCR ERL indexing begins at zero.

Q4. Test valid and invalid inputs. Compare actual and expected results to find faults. A few successful tests cannot prove there are no faults.

I Do / We Do pages · Guided practice

Q1. Retest -1, 0, 100 and 101.

You Do pages · Code First

```
octet = 300
if octet >= 0 AND octet <= 255 then
    print("Valid")
else
    print("Invalid")
endif
```

Expected result: Invalid

You Do pages · Independent questions

Q1. Syntax violates language rules; logic produces incorrect behaviour even if program runs.

Q2. Use AND, not OR; OR incorrectly accepts values outside the range.

Q3. To verify the correction and ensure no new faults were introduced.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: